



PAC Learning Theory

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Learning Theory

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- If the hypothesis space is very complex, can we even find the best h ?
- How complex should h be?
- How do we avoid **overfitting**?

Learning Theory

How many examples are needed for learning?

Learning Theory: Intuition

- Any hypothesis that is *seriously wrong* will almost certainly be “*found out*” with *high probability* after a *small number of examples*, because it will make an incorrect prediction.

Learning Theory: Intuition

- Thus, any hypothesis that is **consistent** with a **sufficiently large** set of training examples is **unlikely** to be **seriously wrong**:
- Such a hypothesis must be **probably approximately correct (PAC)**

PAC Learning

- Any learning algorithm that returns hypotheses that are probably approximately correct is called a **PAC learning algorithm**

PAC Learning: Assumptions

- **Stationarity (Assumption 1):** Future examples are going to be drawn from the same fixed distribution $P(E)=P(X, Y)$ as past examples
- **Noiseless (Assumption 2):** The true function f is deterministic
- **Realizability (Assumption 3):** True function f is a member of the hypothesis space H that is being considered.

PAC Learning: Boolean Functions

- 0/1 Loss:

$$L_{0/1}(y, \hat{y}) = 0 \text{ if } y = \hat{y}, \text{ else } 1$$

- Expected Generalization Loss:

$$GenLoss_L(h) = \sum_{(x,y) \in \mathcal{E}} L(y, h(x)) P(x, y) ,$$

PAC Learning: Boolean Functions

- Expected Generalization Loss error with 0/1 loss:

$$\text{error}(h) = \text{GenLoss}_{L_{0/1}}(h) = \sum_{x,y} L_{0/1}(y, h(x)) P(x, y)$$

- error(h)* is the probability that *h* misclassifies a new example
- A hypothesis *h* is **approximately correct** if *error(h)* $\leq \epsilon$, ϵ is a small constant

PAC Learning: Boolean Functions

Find an N such that,

- after training on N examples,
- with **high probability**,
- all consistent

hypotheses will be approximately correct.

Approximately Correct means:

- Hypothesis h is “close” to the true function f in hypothesis space
- h lies inside what is called the ϵ -ball around the true function f
- The hypothesis space outside ϵ -ball is called $\mathcal{H}_{\text{bad}}$

PAC Learning: Boolean Functions

Derive the probability the a hypothesis $h_b \in \mathcal{H}_{\text{bad}}$ and consistent with first N examples:

- $\text{error}(h_b) > \epsilon$
- Probability (h_b is consistent with a example) $\leq (1 - \epsilon)$
- $P(h_b \text{ is consistent with } N \text{ examples}) \leq (1 - \epsilon)^N$

$$P(\mathcal{H}_{\text{bad}} \text{ contains a consistent hypothesis}) \leq |\mathcal{H}_{\text{bad}}|(1 - \epsilon)^N \leq |\mathcal{H}|(1 - \epsilon)^N$$

Can you prove this inequality?

PAC Learning: Boolean Functions

Let us assume a small probability δ such that

$$P(\mathcal{H}_{\text{bad}} \text{ contains a consistent hypothesis}) \leq |\mathcal{H}|(1 - \epsilon)^N \leq \delta$$

Given that $1 - \epsilon \leq e^{-\epsilon}$, prove the following...

$$N \geq \frac{1}{\epsilon} \left(\ln \frac{1}{\delta} + \ln |\mathcal{H}| \right)$$

Now, calculate the sample complexity (N) for a set of all Boolean functions with n attributes.

PAC Learnability

DEFINITION 3.1 (PAC Learnability) A hypothesis class $\mathcal{H}$ is PAC learnable if there exist a function $m_{\mathcal{H}} : (0, 1)^2 \rightarrow \mathbb{N}$ and a learning algorithm with the following property: For every $\epsilon, \delta \in (0, 1)$, for every distribution $\mathcal{D}$ over $\mathcal{X}$, and for every labeling function $f : \mathcal{X} \rightarrow \{0, 1\}$, if the realizable assumption holds with respect to $\mathcal{H}, \mathcal{D}, f$, then when running the learning algorithm on $m \geq m_{\mathcal{H}}(\epsilon, \delta)$ i.i.d. examples generated by $\mathcal{D}$ and labeled by f , the algorithm returns a hypothesis h such that, with probability of at least $1 - \delta$ (over the choice of the examples), $L_{(\mathcal{D}, f)}(h) \leq \epsilon$.

PAC Learnability

Sample Complexity

The function $m_{\mathcal{H}} : (0, 1)^2 \rightarrow \mathbb{N}$ determines the *sample complexity* of learning $\mathcal{H}$: that is, how many examples are required to guarantee a probably approximately correct solution. The sample complexity is a function of the accuracy (ϵ) and confidence (δ) parameters. It also depends on properties of the hypothesis class $\mathcal{H}$ – for example, for a finite class we showed that the sample complexity depends on $\log$ the size of $\mathcal{H}$.

COROLLARY 3.2 *Every finite hypothesis class is PAC learnable with sample complexity*

$$m_{\mathcal{H}}(\epsilon, \delta) \leq \left\lceil \frac{\log(|\mathcal{H}|/\delta)}{\epsilon} \right\rceil.$$

PAC Learnability: Agnostic Case

Removing the Realizability Assumption

We have required that the learning algorithm succeeds on a pair of data distribution $\mathcal{D}$ and labeling function f provided that the realizability assumption is met. For practical learning tasks, this assumption may be too strong (can we really guarantee that there is a rectangle in the color-hardness space that *fully determines* which papayas are tasty?). In the next subsection, we will describe the *agnostic PAC* model in which this realizability assumption is waived.

PAC Learnability: Agnostic Case

Learning Problems beyond Binary Classification

The learning task that we have been discussing so far has to do with predicting a binary label to a given example (like being tasty or not). However, many learning tasks take a different form. For example, one may wish to predict a real valued number (say, the temperature at 9:00 p.m. tomorrow) or a label picked from a finite set of labels (like the topic of the main story in tomorrow's paper). It turns out that our analysis of learning can be readily extended to such and many other scenarios by allowing a variety of loss functions. We shall discuss that in Section 3.2.2 later.

PAC Learnability: Agnostic Case

A More Realistic Model for the Data-Generating Distribution

Recall that the realizability assumption requires that there exists $h^* \in \mathcal{H}$ such that $\mathbb{P}_{x \sim \mathcal{D}}[h^*(x) = f(x)] = 1$. In many practical problems this assumption does not hold. Furthermore, it is maybe more realistic not to assume that the labels are fully determined by the features we measure on input elements (in the case of the papayas, it is plausible that two papayas of the same color and softness will have different taste). In the following, we relax the realizability assumption by replacing the “target labeling function” with a more flexible notion, a data-labels generating distribution.

PAC Learnability: Agnostic Case

- A More Realistic Model for the Data-Generating Distribution

Formally, from now on, let $\mathcal{D}$ be a probability distribution over $\mathcal{X} \times \mathcal{Y}$, where, as before, $\mathcal{X}$ is our domain set and $\mathcal{Y}$ is a set of labels (usually we will consider $\mathcal{Y} = \{0, 1\}$). That is, $\mathcal{D}$ is a *joint distribution* over domain points and labels. One can view such a distribution as being composed of two parts: a distribution $\mathcal{D}_x$ over unlabeled domain points (sometimes called the *marginal distribution*) and a *conditional* probability over labels for each domain point, $\mathcal{D}((x, y)|x)$. In the papaya example, $\mathcal{D}_x$ determines the probability of encountering a papaya whose color and hardness fall in some color-hardness values domain, and the conditional probability is the probability that a papaya with color and hardness represented by x is tasty. Indeed, such modeling allows for two papayas that share the same color and hardness to belong to different taste categories.

PAC Learnability: Agnostic Case

- A More Realistic Model for the Data-Generating Distribution

The empirical and the True Error Revised

For a probability distribution, $\mathcal{D}$, over $\mathcal{X} \times \mathcal{Y}$, one can measure how likely h is to make an error when labeled points are randomly drawn according to $\mathcal{D}$. We redefine the true error (or risk) of a prediction rule h to be

$$L_{\mathcal{D}}(h) \stackrel{\text{def}}{=} \mathbb{P}_{(x,y) \sim \mathcal{D}}[h(x) \neq y] \stackrel{\text{def}}{=} \mathcal{D}(\{(x, y) : h(x) \neq y\}). \quad (3.1)$$

$$L_S(h) \stackrel{\text{def}}{=} \frac{|\{i \in [m] : h(x_i) \neq y_i\}|}{m}.$$

Given S , a learner can compute $L_S(h)$ for any function $h : X \rightarrow \{0, 1\}$. Note that $L_S(h) = L_{D(\text{uniform over } S)}(h)$.

PAC Learnability: Agnostic Case

The Goal

We wish to find some hypothesis, $h : \mathcal{X} \rightarrow \mathcal{Y}$, that (probably approximately) minimizes the true risk, $L_D(h)$.

The Bayes Optimal Predictor.

Given any probability distribution $\mathcal{D}$ over $\mathcal{X} \times \{0, 1\}$, the best label predicting function from $\mathcal{X}$ to $\{0, 1\}$ will be

$$f_{\mathcal{D}}(x) = \begin{cases} 1 & \text{if } \mathbb{P}[y = 1|x] \geq 1/2 \\ 0 & \text{otherwise} \end{cases}$$

It is easy to verify (see Exercise 7) that for every probability distribution $\mathcal{D}$, the Bayes optimal predictor $f_{\mathcal{D}}$ is optimal, in the sense that no other classifier, $g : \mathcal{X} \rightarrow \{0, 1\}$ has a lower error. That is, for every classifier g , $L_{\mathcal{D}}(f_{\mathcal{D}}) \leq L_{\mathcal{D}}(g)$.

PAC Learnability: Agnostic Case

The Bayes Optimal Predictor.

Given any probability distribution $\mathcal{D}$ over $\mathcal{X} \times \{0, 1\}$, the best label predicting function from $\mathcal{X}$ to $\{0, 1\}$ will be

$$f_{\mathcal{D}}(x) = \begin{cases} 1 & \text{if } \mathbb{P}[y = 1|x] \geq 1/2 \\ 0 & \text{otherwise} \end{cases}$$

Clearly, we cannot hope that the learning algorithm will find a hypothesis whose error is smaller than the minimal possible error, that of the Bayes predictor.

PAC Learnability: Agnostic Case

DEFINITION 3.3 (Agnostic PAC Learnability) A hypothesis class $\mathcal{H}$ is agnostic PAC learnable if there exist a function $m_{\mathcal{H}} : (0, 1)^2 \rightarrow \mathbb{N}$ and a learning algorithm with the following property: For every $\epsilon, \delta \in (0, 1)$ and for every distribution $\mathcal{D}$ over $\mathcal{X} \times \mathcal{Y}$, when running the learning algorithm on $m \geq m_{\mathcal{H}}(\epsilon, \delta)$ i.i.d. examples generated by $\mathcal{D}$, the algorithm returns a hypothesis h such that, with probability of at least $1 - \delta$ (over the choice of the m training examples),

$$L_{\mathcal{D}}(h) \leq \min_{h' \in \mathcal{H}} L_{\mathcal{D}}(h') + \epsilon.$$

Generalized learning problem

Multiclass classification, Regression

Generalized Loss Functions

Given any set $\mathcal{H}$ (that plays the role of our hypotheses, or models) and some domain Z let ℓ be any function from $\mathcal{H} \times Z$ to the set of nonnegative real numbers, $\ell : \mathcal{H} \times Z \rightarrow \mathbb{R}_+$.

We call such functions *loss functions*.

Generalized learning problem

Multiclass classification, Regression

We now define the *risk function* to be the expected loss of a classifier, $h \in \mathcal{H}$, with respect to a probability distribution D over Z , namely,

$$L_{\mathcal{D}}(h) \stackrel{\text{def}}{=} \mathbb{E}_{z \sim \mathcal{D}} [\ell(h, z)]. \quad (3.3)$$

That is, we consider the expectation of the loss of h over objects z picked randomly according to $\mathcal{D}$. Similarly, we define the *empirical risk* to be the expected loss over a given sample $S = (z_1, \dots, z_m) \in Z^m$, namely,

$$L_S(h) \stackrel{\text{def}}{=} \frac{1}{m} \sum_{i=1}^m \ell(h, z_i). \quad (3.4)$$

Generalized learning problem

Multiclass classification, Regression

- **0–1 loss:** Here, our random variable z ranges over the set of pairs $\mathcal{X} \times \mathcal{Y}$ and the loss function is

$$\ell_{0-1}(h, (x, y)) \stackrel{\text{def}}{=} \begin{cases} 0 & \text{if } h(x) = y \\ 1 & \text{if } h(x) \neq y \end{cases}$$

This loss function is used in binary or multiclass classification problems.

- **Square Loss:** Here, our random variable z ranges over the set of pairs $\mathcal{X} \times \mathcal{Y}$ and the loss function is

$$\ell_{\text{sq}}(h, (x, y)) \stackrel{\text{def}}{=} (h(x) - y)^2.$$

Generalized learning problem

Multiclass classification, Regression

DEFINITION 3.4 (Agnostic PAC Learnability for General Loss Functions) A hypothesis class $\mathcal{H}$ is agnostic PAC learnable with respect to a set Z and a loss function $\ell : \mathcal{H} \times Z \rightarrow \mathbb{R}_+$, if there exist a function $m_{\mathcal{H}} : (0, 1)^2 \rightarrow \mathbb{N}$ and a learning algorithm with the following property: For every $\epsilon, \delta \in (0, 1)$ and for every distribution $\mathcal{D}$ over Z , when running the learning algorithm on $m \geq m_{\mathcal{H}}(\epsilon, \delta)$ i.i.d. examples generated by $\mathcal{D}$, the algorithm returns $h \in \mathcal{H}$ such that, with probability of at least $1 - \delta$ (over the choice of the m training examples),

$$L_{\mathcal{D}}(h) \leq \min_{h' \in \mathcal{H}} L_{\mathcal{D}}(h') + \epsilon,$$

where $L_{\mathcal{D}}(h) = \mathbb{E}_{z \sim \mathcal{D}}[\ell(h, z)]$.

Some Remarks

Remark 3.1 (A Note About Measurability*) In the aforementioned definition, for every $h \in \mathcal{H}$, we view the function $\ell(h, \cdot) : Z \rightarrow \mathbb{R}_+$ as a random variable and define $L_{\mathcal{D}}(h)$ to be the expected value of this random variable. For that, we need to require that the function $\ell(h, \cdot)$ is measurable. Formally, we assume that there is a σ -algebra of subsets of Z , over which the probability $\mathcal{D}$ is defined, and that the preimage of every initial segment in $\mathbb{R}_+$ is in this σ -algebra. In the specific case of binary classification with the 0–1 loss, the σ -algebra is over $\mathcal{X} \times \{0, 1\}$ and our assumption on ℓ is equivalent to the assumption that for every h , the set $\{(x, h(x)) : x \in \mathcal{X}\}$ is in the σ -algebra.

Some Remarks

Remark 3.2 (Proper versus Representation-Independent Learning*) In the preceding definition, we required that the algorithm will return a hypothesis from $\mathcal{H}$. In some situations, $\mathcal{H}$ is a subset of a set $\mathcal{H}'$, and the loss function can be naturally extended to be a function from $\mathcal{H}' \times Z$ to the reals. In this case, we may allow the algorithm to return a hypothesis $h' \in \mathcal{H}'$, as long as it satisfies the requirement $L_{\mathcal{D}}(h') \leq \min_{h \in \mathcal{H}} L_{\mathcal{D}}(h) + \epsilon$. Allowing the algorithm to output a hypothesis from $\mathcal{H}'$ is called *representation independent* learning, while proper learning occurs when the algorithm must output a hypothesis from $\mathcal{H}$. Representation independent learning is sometimes called “improper learning,” although there is nothing improper in representation independent learning.

Exercise

1. **Monotonicity of Sample Complexity:** Let $\mathcal{H}$ be a hypothesis class for a binary classification task. Suppose that $\mathcal{H}$ is PAC learnable and its sample complexity is given by $m_{\mathcal{H}}(\cdot, \cdot)$. Show that $m_{\mathcal{H}}$ is monotonically nonincreasing in each of its parameters. That is, show that given $\delta \in (0, 1)$, and given $0 < \epsilon_1 \leq \epsilon_2 < 1$, we have that $m_{\mathcal{H}}(\epsilon_1, \delta) \geq m_{\mathcal{H}}(\epsilon_2, \delta)$. Similarly, show that given $\epsilon \in (0, 1)$, and given $0 < \delta_1 \leq \delta_2 < 1$, we have that $m_{\mathcal{H}}(\epsilon, \delta_1) \geq m_{\mathcal{H}}(\epsilon, \delta_2)$.

Exercise

2. Let $\mathcal{X}$ be a discrete domain, and let $\mathcal{H}_{\text{Singleton}} = \{h_z : z \in \mathcal{X}\} \cup \{h^-\}$, where for each $z \in \mathcal{X}$, h_z is the function defined by $h_z(x) = 1$ if $x = z$ and $h_z(x) = 0$ if $x \neq z$. h^- is simply the all-negative hypothesis, namely, $\forall x \in X, h^-(x) = 0$. The realizability assumption here implies that the true hypothesis f labels negatively all examples in the domain, perhaps except one.
1. Describe an algorithm that implements the ERM rule for learning $\mathcal{H}_{\text{Singleton}}$ in the realizable setup.
 2. Show that $\mathcal{H}_{\text{Singleton}}$ is PAC learnable. Provide an upper bound on the sample complexity.

Exercise

3. Let $\mathcal{X} = \mathbb{R}^2$, $\mathcal{Y} = \{0, 1\}$, and let $\mathcal{H}$ be the class of concentric circles in the plane, that is, $\mathcal{H} = \{h_r : r \in \mathbb{R}_+\}$, where $h_r(x) = \mathbb{1}_{[\|x\| \leq r]}$. Prove that $\mathcal{H}$ is PAC learnable (assume realizability), and its sample complexity is bounded by

$$m_{\mathcal{H}}(\epsilon, \delta) \leq \left\lceil \frac{\log(1/\delta)}{\epsilon} \right\rceil.$$

Exercise

7. (*) **The Bayes optimal predictor:** Show that for every probability distribution $\mathcal{D}$, the Bayes optimal predictor $f_{\mathcal{D}}$ is optimal, in the sense that for every classifier g from $\mathcal{X}$ to $\{0, 1\}$, $L_{\mathcal{D}}(f_{\mathcal{D}}) \leq L_{\mathcal{D}}(g)$.

Implications of PAC on generalization

If the hypothesis space contains enough hypothesis to classify the given set of example all possible ways –

- Sample complexity in PAC requires all possible examples

So we may restrict the hypothesis space –

- It may miss the true function

To escape this dilemma –

- Prior knowledge, Ockham's razor, learnable subsets of the entire hypothesis space

Acknowledgement

- Chapter 3: Shalev-Shwartz, S., & Ben-David, S. (2014). Understanding machine learning: From theory to algorithms. Cambridge University Press
- Chapter 18 (3rd ed.)/19(4th ed.) (AIAMA) (Textbook)
- Online resources including LLMs

Feedback

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